

Key Features and Properties of Parabolas

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A **parabola** is the graph of a quadratic function,
usually written as $y = ax^2 + bx + c$ or
 $y = a(x - h)^2 + k$.

Vertex

The **vertex** is the highest or lowest point of the parabola.

In the vertex form

$$y = a(x - h)^2 + k$$

the vertex is (h, k) .

Axis of Symmetry

The **axis of symmetry** is the vertical line passing through the vertex.

Its equation is

$$x = h$$

or

$$x = -\frac{b}{2a}.$$

Opening Direction

The **opening direction** depends on the value of a .

If $a > 0$, the parabola opens upward.

If $a < 0$, the parabola opens downward.

x -Intercepts and Zeros

The x -**intercepts** or **zeros** are the points where the graph crosses the x -axis.

They are found by setting

$$y = 0.$$

y -Intercept

The y -**intercept** is the point where the graph crosses the y -axis.

It is found by letting

$$x = 0.$$

Minimum and Maximum

If the parabola opens upward, the vertex gives the **minimum value**.

If it opens downward, the vertex gives the **maximum value**.

Increasing and Decreasing

If a parabola opens upward, it is **decreasing** to the left of the vertex and **increasing** to the right.

If it opens downward, it is **increasing** to the left and **decreasing** to the right.

Examples

Example 1

Given

$$y = (x - 2)^2 - 5$$

determine the vertex, axis of symmetry, opening direction, and minimum or maximum value.

Example 2

For

$$y = -2(x + 1)^2 + 8$$

identify the vertex, axis of symmetry, opening direction, and minimum or maximum value.

Example 3

Find the x -intercepts, y -intercept, and zeros of

$$y = x^2 - 5x + 6.$$

Example 4

Find the x -intercepts, y -intercept, and zeros of

$$y = x^2 - 9.$$

Example 5

State whether

$$y = (x + 4)^2 - 3$$

is increasing or decreasing on $(-\infty, -4)$ and $(-4, \infty)$.

Example 6

State whether

$$y = -(x - 5)^2 + 7$$

is increasing or decreasing on $(-\infty, 5)$ and $(5, \infty)$.

Example 7

Complete the table for

$$y = x^2 + 4x + 3.$$

Vertex	
Axis of symmetry	
Opening direction	
x -intercepts	
y -intercept	
Minimum/Maximum value	

Example 8

Complete the table for

$$y = -x^2 + 6x - 5.$$

Vertex	
Axis of symmetry	
Opening direction	
x -intercepts	
y -intercept	
Minimum/Maximum value	

Example 9

A ball follows the path

$$h = -x^2 + 8x.$$

Determine the maximum height, the horizontal position where the maximum height occurs, and the zeros of the function.

Example 10

The height of a fountain is modeled by

$$h = -2(x - 3)^2 + 18.$$

Identify the vertex, axis of symmetry, opening direction, and maximum value.

Practice Exercises

Practice 1

Given

$$y = (x - 2)^2 - 5$$

determine the vertex, axis of symmetry, opening direction, and minimum or maximum value.

Practice 2

For

$$y = -2(x + 1)^2 + 8$$

identify the vertex, axis of symmetry, opening direction, and minimum or maximum value.

Practice 3

Find the x -intercepts, y -intercept, and zeros of

$$y = x^2 - 5x + 6.$$

Practice 4

Find the x -intercepts, y -intercept, and zeros of

$$y = x^2 - 9.$$

Practice 5

State whether

$$y = (x + 4)^2 - 3$$

is increasing or decreasing on $(-\infty, -4)$ and $(-4, \infty)$.

Practice 6

State whether

$$y = -(x - 5)^2 + 7$$

is increasing or decreasing on $(-\infty, 5)$ and $(5, \infty)$.

Practice 7

Complete the table for

$$y = x^2 + 4x + 3.$$

Vertex	
Axis of symmetry	
Opening direction	
x -intercepts	
y -intercept	
Minimum/Maximum value	

Practice 8

Complete the table for

$$y = -x^2 + 6x - 5.$$

Vertex	
Axis of symmetry	
Opening direction	
x -intercepts	
y -intercept	
Minimum/Maximum value	

Practice 9

A ball follows the path

$$h = -x^2 + 8x.$$

Determine the maximum height, the horizontal position where the maximum height occurs, and the zeros of the function.

Practice 10

The height of a fountain is modeled by

$$h = -2(x - 3)^2 + 18.$$

Identify the vertex, axis of symmetry, opening direction, and maximum value.

Activities

Activity 1

Given

$$y = (x + 3)^2 - 6$$

determine the vertex, axis of symmetry, opening direction, and minimum or maximum value.

Activity 2

For

$$y = -3(x - 2)^2 + 12$$

identify the vertex, axis of symmetry, opening direction, and minimum or maximum value.

Activity 3

Find the x -intercepts, y -intercept, and zeros of

$$y = x^2 - 7x + 12.$$

Activity 4

Find the x -intercepts, y -intercept, and zeros of

$$y = x^2 - 16.$$

Activity 5

State whether

$$y = (x - 1)^2 + 4$$

is increasing or decreasing on $(-\infty, 1)$ and $(1, \infty)$.

Activity 6

State whether

$$y = -(x + 2)^2 + 9$$

is increasing or decreasing on $(-\infty, -2)$ and $(-2, \infty)$.

Activity 7

Complete the table for

$$y = x^2 - 6x + 8.$$

Vertex	
Axis of symmetry	
Opening direction	
x -intercepts	
y -intercept	
Minimum/Maximum value	

Activity 8

Complete the table for

$$y = -x^2 + 4x + 5.$$

Vertex	
Axis of symmetry	
Opening direction	
x -intercepts	
y -intercept	
Minimum/Maximum value	

Activity 9

A ball follows the path

$$h = -x^2 + 10x.$$

Determine the maximum height, the horizontal position where the maximum height occurs, and the zeros of the function.

Activity 10

The height of a fountain is modeled by

$$h = -3(x - 2)^2 + 27.$$

Identify the vertex, axis of symmetry, opening direction, and maximum value.

Thank you!